

Technical Appendix: Genomic Privacy and Secure Health Data Systems

Subject: Technical Feasibility of Privacy-Preserving Architectures for Genomic and Health Data

Context: Supporting Amicus Brief for SCOTUS Review of Genetic Information and Disability Rights

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1. Introduction

This appendix provides a technical summary of the genomic findings and privacy-enhancing technology (PET) solutions integrated within the PhD portfolio. It demonstrates that robust, verifiable privacy protections are technically feasible and should be considered a “least restrictive alternative” in legal and policy frameworks concerning sensitive health data.

2. Genomic Findings and Methodology

The portfolio analyzed a 23andMe dataset (genomeCaustinMcLaughlinFull20131024071559.txt) comprising approximately 960,000 variants. Key validated findings include:

Metric	Finding	Significance
Biological Sex	X-chromosome heterozygosity 0.57%	Confirms male XY status; establishes baseline for chromosomal integrity.
Heterozygosity	Autosomal heterozygosity 29.38%	Demonstrates standard genetic diversity and array reliability.
ROH	1,504 segments (avg ~82 SNPs)	Identifies runs of homozygosity, relevant for recessive trait analysis.
GC Bias	~52.85% (array-driven)	Highlights the necessity of accounting for technical bias in genomic modeling.
HLA Region	~3.9× higher SNP density	Confirms high-resolution mapping of critical immune system regions.

Methodology: Analysis was performed using custom Python scripts (Pandas, NumPy) for statistical validation, including Kolmogorov-Smirnov (KS) tests and Z-score normalization to ensure data reliability.

3. Privacy-Enhancing Technology (PET) Solutions

The portfolio proposes a multi-layered technical stack to protect genomic privacy while enabling clinical and secondary research use:

A. Trusted Execution Environments (TEEs)

TEEs (e.g., ARM TrustZone, Intel SGX) provide hardware-enforced isolation for sensitive computations.

- **Application:** Genomic queries (e.g., BRCA status, PRS calculation) are executed within an encrypted enclave.
- **Privacy Gain:** Raw genomic data is never exposed to the host operating system or unauthorized users, mitigating re-identification risks.

B. Blockchain + TEE Hybrids

Integrating blockchain with TEEs (e.g., Aquareum, Golden Grain) creates an immutable, auditable record of data access.

- **Application:** Every access request and computation result is logged on a tamper-proof ledger.
- **Privacy Gain:** Ensures verifiable compliance with data-sharing agreements and provides a clear audit trail for regulatory oversight (HIPAA/GINA).

C. Privacy-Preserving GWAS and Machine Learning

Techniques such as Federated Learning and Differential Privacy enable research on aggregate data without sharing individual records.

- **Application:** Training ADHD risk models across multiple health systems without centralizing sensitive genomic data.
- **Privacy Gain:** Mathematically guarantees that individual contributions to a dataset cannot be reverse-engineered.

4. Human Cost of Adjudication Delays

Data from the Social Security Administration (FY2023) highlights the urgency of implementing these technical solutions. A total of **33,270** individuals died while their disability claims were pending.

State	Deaths While Pending (FY2023)
Texas	3,470
Florida	3,070
California	2,230
Georgia	2,050

Conclusion: The integration of secure, genomic-informed risk stratification (e.g., via pathway PRSs) could significantly expedite these critical reviews, upholding procedural due process as established in *Mathews v. Eldridge*.

5. Final Recommendations

1. **Judicial Recognition:** Courts should recognize TEE and blockchain solutions as technically feasible measures for protecting genetic privacy.
 2. **Regulatory Mandate:** Federal agencies (SSA, HHS) should adopt these architectures to secure health data sharing and expedite disability adjudication.
 3. **Future-Proofing:** Policies must account for post-quantum cryptographic readiness to protect genomic data assets that will remain sensitive for decades.
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References

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6. *Dobbs v. Jackson Women's Health Organization*, 597 U.S. ____ (2022).
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8. Health Insurance Portability and Accountability Act (HIPAA) of 1996.